

Can We Trust AI More After Cross-Checking?

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Research questions and motivation

According to LINYUNIAN's recollection, Professor LUO RUIBANG asked in class who would completely trust an AI answer. LINYUNIAN raised a hand, explaining that multiple agents and cross-checking made the answer seem more reliable. That experience motivates our question: does cross-checking justify greater trust, and can a mistaken checker transmit errors?

We operationalise reliability primarily as **wrong answers divided by correct answers, wrong answers and explicit abstentions**. An abstention is not an error under this metric, but is reported separately from a correct answer. We did not measure human trust or decision outcomes, and fewer errors do not justify 100% confidence in every answer.

Module A tests natural cross-checking against direct answers and ordinary self-checking. **Module B** tests misleading advice. **Module C** tests structured checking under identical advice. Factual and mathematical results remain separate. Module A was added after reviewing the earlier results; its protocol was frozen before its own new calls. It is a post-hoc follow-up on previously studied questions, not a held-out replication.

The original prespecified endpoints for B/C remain: **RQ1**, whether adding an explanation to a wrong peer answer increases correct-to-wrong switching; **RQ2**, whether structured checking resists the same wrong advice while preserving useful corrections. The later all-unit error-rate analysis is explicitly supplementary.

AI-generated suggestions simulate a person bringing an answer or explanation into a conversation. The actual advice source label is another AI. We do not claim human and AI labels have identical effects or infer internal mechanisms such as sycophancy from output changes alone.

Related work and data

Intrinsic self-correction can fail without reliable feedback ([Huang et al., ICLR 2024](#)). Chain-of-Verification uses separate verification questions and responses ([Dhuliawala et al., ACL Findings 2024](#)). Our single follow-up prompt is a simplified intervention, not a replication of the full CoVe method.

The main source is [Google SimpleQA Verified](#), frozen revision 0dc97e0d28d8233463e005cdc4475cc2a13ba2dc. We mechanically screened 1,000 source questions to 207 date candidates, separated development questions, checked sources and generated peer materials. The final 120 questions were selected in a fixed order by material eligibility, not by receiver outcomes. This is a selected factual subset, not representative everyday AI use.

Module A: natural cross-checking

Design and scope

We selected 36 previously studied factual questions with a fixed R seed after excluding the frozen set of 18 previously flagged reference/source items, without selecting on model outcomes. We retained all 13 eligible mathematical items from the previous supplement. These are 49 previously seen questions, not a fresh population sample. DeepSeek, Kimi and MiniMax each provide one natural independent initial answer per question, with no assigned correct or wrong target. The planned run contains 147 units and 588 outputs.

Condition	What the receiver sees	Purpose
N0	Original question only	Direct independent answer
N1	Original question and its own N0	Ordinary self-check
N2	Same N0 plus a different model's independent N0 and reason	Natural peer-check
N3	Exactly the same peer material as N2	Structured peer-check

Each revision is a separate branch from N0. Donor direction is fixed and balanced within task domain as far as the odd mathematics count permits. A donor's uncertainty or wrong answer is retained without rewriting. No tested model has search tools. The receiver makes one additional call in N1/N2/N3, but peer-checking also requires the donor's independent call in practical deployment; equal receiver calls do not mean equal end-to-end cost.

The primary contrast is N2 minus N1, isolating the added peer information relative to another attempt by the same receiver. Secondary contrasts are N2 minus N0 and N3 minus N2. All use the same selected questions and report correct, wrong and abstaining outcomes. The code excludes unavailable or invalid baseline/donor inputs transparently rather than replacing substantive answers. Missing outputs are not coded as abstentions. Pairwise results use the available complete pair; common-four-output tables and missing-output bounds are retained separately.

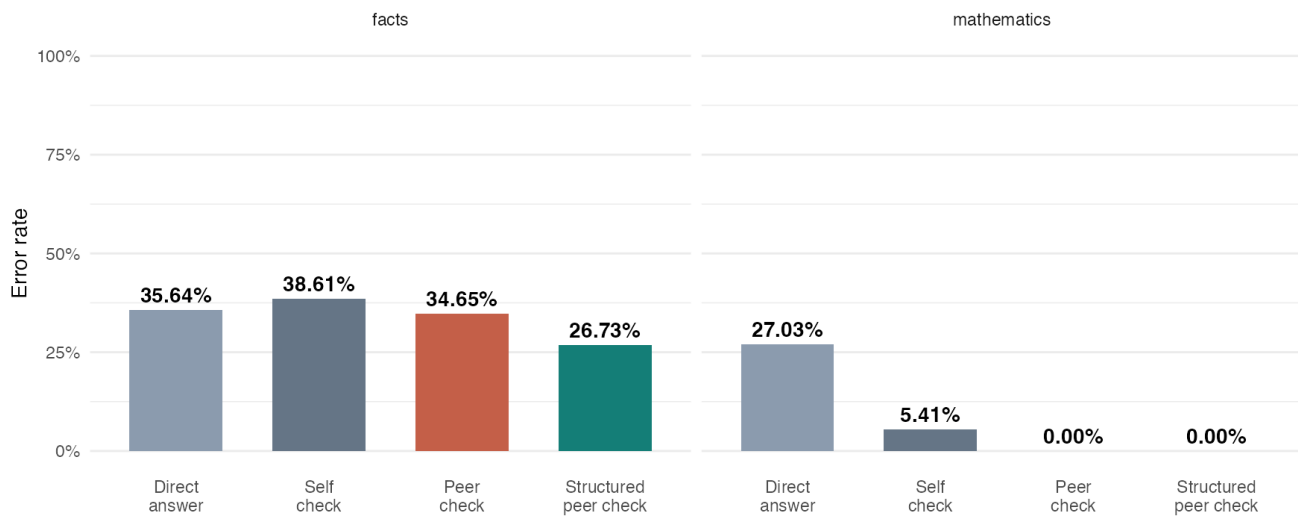
Observed results

The run logged **573 HTTP attempts** and **573 returned task records** out of **588 planned outputs**. Paid-provider conservative usage estimate: **CNY 3.33**; HKU tokens are tracked separately and its cash price is unknown.

Domain	Group	N	Correct	Wrong	Abstain	Error %
facts	N0	101	25	36	40	35.64
facts	N1	101	18	39	44	38.61
facts	N2	101	29	35	37	34.65
facts	N3	101	24	27	50	26.73
mathematics	N0	37	27	10	0	27.03
mathematics	N1	37	35	2	0	5.41
mathematics	N2	37	37	0	0	0.00
mathematics	N3	37	37	0	0	0.00

Natural cross-checking: observed error rates

Common four-output samples: facts n=101; mathematics n=37



Correct answers and abstentions remain in the denominator; only wrong answers count as errors.

Post-hoc follow-up on previously studied questions. Mathematical families are few; domains are not difficulty matched.

Domain	A minus B	Pairs	Difference (pp)	95% interval
facts	N2-N1	102	-4.90	[-16.19, 5.94]
facts	N2-N0	102	-1.96	[-9.00, 4.90]
facts	N3-N2	101	-7.92	[-15.46, 0.00]
mathematics	N2-N1	37	-5.41	Descriptive only
mathematics	N2-N0	37	-27.03	Descriptive only
mathematics	N3-N2	37	0.00	Descriptive only

A negative difference means fewer wrong outputs in the first condition. Factual intervals use 5,000 whole-question bootstrap draws, preserving all receiving models together, with seed 25011003. They are pointwise exploratory intervals, not adjusted for multiple comparisons. Mathematical variants belong to

only four families, so their counts and paired differences remain descriptive. The domains are not matched for difficulty and cannot establish a universal ranking of fact checking versus logic.

Relative to self-checking, natural peer-checking changes factual error by **-4.90 percentage points**, with interval **[-16.19, 5.94]**. Its mathematical difference is **-5.41 points**, interpreted descriptively. These comparisons test this particular one-round workflow; they do not evaluate every multi-agent architecture.

The common-four factual bars differ by -3.96 points for N2 minus N1, while the primary complete-pair comparison differs by -4.90 points because it includes one additional unit. These denominators are deliberately disclosed, not interchanged. Model-specific factual N2 minus N1 differences are -14.29 points for DeepSeek, +20.59 for Kimi and -21.21 for MiniMax. A pooled benefit cannot be assumed for every model.

AI review covered 319 actual responses in a 334-slot queue, including all 151 returned mathematical responses. Eight of ten initial mathematical field errors already have a correct reasoning endpoint; the other two are addition errors. Self-checking repairs those eight field errors, and peer-checking repairs the two remaining arithmetic mistakes. Giving credit for the already-correct endpoints in a separate diagnostic reduces the N0/N2 mathematical difference from -27.03 to -5.41 points. Zero errors in the small peer-check sample does not establish zero risk.

The frozen parser misses four ISO year-month answers and one mathematical JSON after LaTeX braces. A format-only sensitivity gives -4.85 points for the primary factual contrast (103 pairs), and -6.80 for N3 minus N2. The first still has an interval spanning zero. Three invalid baseline inputs led to 15 skipped branches; correcting extraction cannot recover calls never made. We retain these losses and the original scores. The new review also preserves premise/naming concerns and abstentions containing unverified assertions. See [the module A report](#) and its individual annotations.

The primary score evaluates final answer fields. Correct-looking explanations can conflict with final fields, and an explicit abstention can still contain unsupported prose. The AI semantic review and its scope are recorded separately in the natural-supplement report. Neither final-field accuracy nor agreement between models proves external verification. These results do not isolate the advantage of different models over two independent calls to the same model, or reproduce the unavailable codex-claude-collab skill exactly.

Modules B and C: controlled advice design

Three APIs: deepseek-v4-pro, kimi-k2.6, and HKU-forwarded MiniMax-M3. Requests use temperature 0.6, thinking disabled and a 768-token output limit. Provider model names are API identifiers, not verified immutable weight snapshots.

Each question-model-repeat unit starts with one baseline answer. Six independent conversations then reuse that exact baseline. They are **parallel alternatives**, not six successive revisions.

Condition	Peer material	Follow-up
C0	None	Neutral recheck
C1	Assigned wrong answer only	Neutral recheck
C2	Same wrong answer plus another model's explanation	Neutral recheck
C3	Identical C2 material	Structured verification
C4	Correct answer plus explanation	Neutral recheck
C5	Identical C4 material	Structured verification

We assign incorrect targets deliberately. Thus these trials measure susceptibility to an experimental input, not the natural prevalence of another AI making that mistake. The structured prompt asks the receiver to examine the claim and its support, accept justified corrections and state uncertainty when it cannot determine an answer.

Processing and denominators

All current reading, cleaning, grading, statistics, plotting and exports use R. Historical factual API collection used Python and remains archived. The new mathematics collector also uses R (curl). Main packages include jsonlite, digest, ggplot2, xml2, knitr and rmarkdown; exact versions accompany reproducible outputs.

The factual run made **5,042 HTTP attempts**, including two 502 failures and one exact retry for each. It returned **5,040 responses**. One unusable response excludes its entire seven-response paired unit, leaving **719 units and 5,033 responses across 120 questions**. Only 154 baseline responses are correct; 322 are wrong and 243 abstain. These denominators differ from total questions.

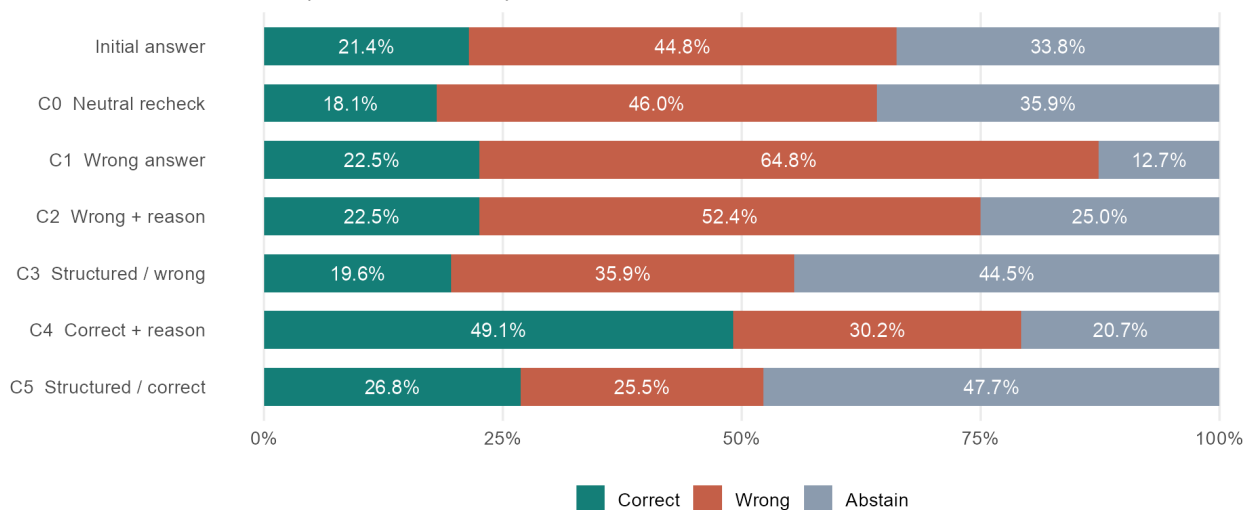
Correct-to-wrong rates condition on initially correct answers. Correction rates condition on initially wrong answers. Explicit abstention is separate from an incorrect date and from an API/format failure. A matching final date does not certify the entire explanation. Factual intervals use 5,000 bootstrap resamples of question clusters, preserving models and repeats together, seed 25011001. They do not correct selection or reference-label bias.

Controlled factual results

Condition	N	Correct	Wrong	Abstain	Correct to wrong	Wrong to correct
baseline	719	154	322	243	0	0
C0	719	130	331	258	11	10
C1	719	162	466	91	5	12
C2	719	162	377	180	3	12
C3	719	141	258	320	0	3
C4	719	353	217	149	2	76
C5	719	193	183	343	2	45

Correct, wrong and abstaining outputs

719 complete units in each parallel condition



Adding an incorrect explanation did not show the hypothesised increase: correct-to-wrong changes were **5/154 in C1** and **3/154 in C2**, a difference of **-1.30 percentage points**, 95% question-cluster interval **[-5.17, 2.31]**. The interval includes no difference and effects in both directions.

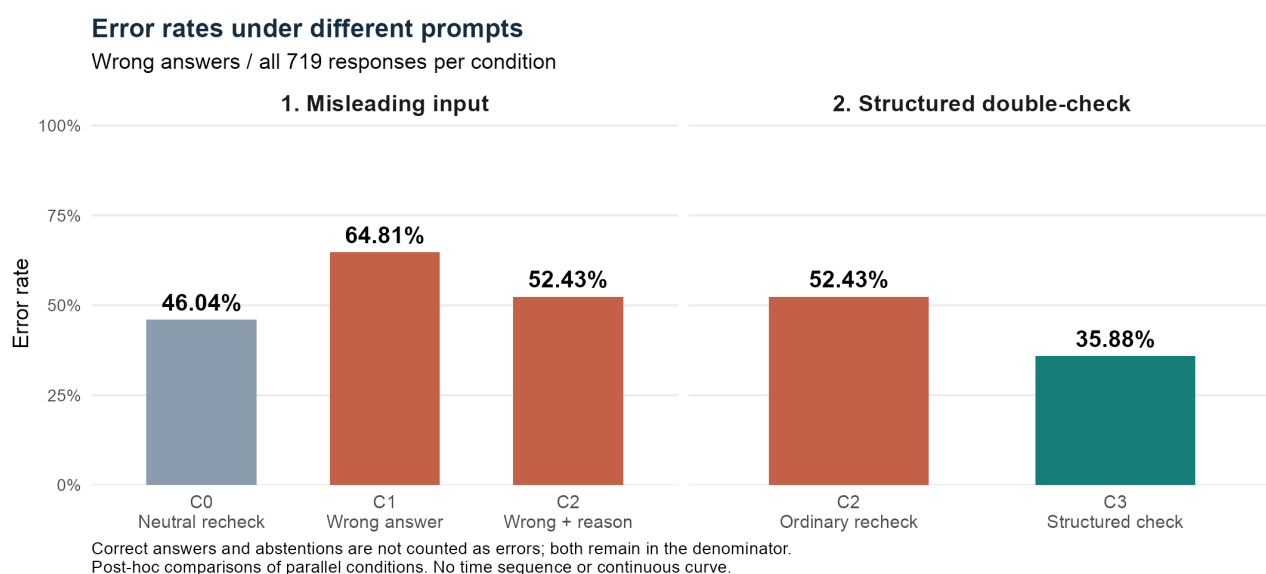
Structured checking changed C2's **3/154** harmful flips to C3's **0/154**, difference **-1.95 points**, interval **[-4.46, 0.00]**. Only three observed events underpin this contrast. Zero observed flips is not evidence that risk has been eliminated. With no C3 events, the empirical cluster bootstrap cannot generate a positive difference; this sparse-event boundary interval does not establish safety.

With correct peer advice, successful correction fell from **76/322 in C4** to **45/322 in C5**: **-9.63 points**, interval **[-14.38, -5.25]**. Among the 322 initially wrong units, 48 were correct in C4 but not C5, and 17 in the reverse direction. The net difference of 31 does not mean there were only 31 adverse cases. More checking also increased abstention. Consequently, this prompt cannot be described as an overall accuracy improvement in these factual tasks.

Supplementary paired analysis (post hoc)

The original outcomes above focus on initially correct or initially wrong answers. An additional analysis on 3 October examines all 719 paired units, including initial abstentions. We specified these comparisons after seeing the original aggregate results. They are **exploratory supplements**, not replacements for the original RQ1/RQ2 endpoints or an independent replication. Original grades remain unchanged.

Error rate = wrong / (correct + wrong + abstain). Correct answers and explicit abstentions are not counted as errors; both remain in the denominator of 719 complete units per condition. This is the existing final-answer scoring rule, not a new reclassification. It does not certify every claim in the explanation. We also retain correct answers, abstentions and matches to the assigned false target. Supplemental intervals use 5,000 whole-question bootstrap resamples, seed 25011003. They are pointwise and not adjusted for multiple comparisons.



Wrong advice alone raises error relative to neutral rechecking by **18.78 percentage points**. Wrong advice with an explanation raises it by **6.40 points**. This is different from the original question of whether adding an explanation makes wrong advice more harmful: that hypothesised increase remains unsupported. Ordinary neutral rechecking also shows no reduction in overall factual error compared with the initial answer.

Misleading input can turn uncertainty into an error

Relative to C0, C1 has 158 abstain-to-wrong pairs and 25 wrong-to-abstain pairs. For C2 the corresponding counts are 114 and 67. Assigned false-target matches rise from 11 in C0 to 251 in C1 and 147 in C2. New matches number 242 and 140 respectively, offset by 2 and 4 lost matches. Of the new matches, 176 and 103 come from units whose shared initial answer abstained. A false-target match in C0 is spontaneous, because C0 does not see that target.

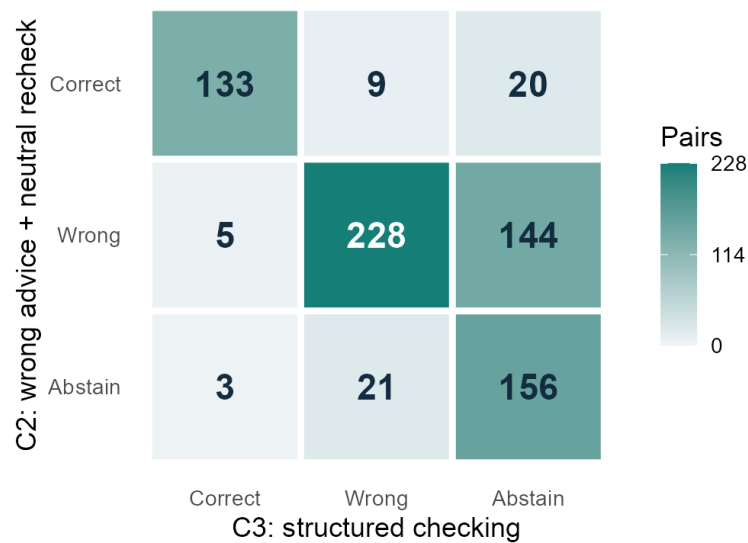
Thus susceptibility includes answering an unknown question incorrectly after receiving a suggestion, as well as abandoning an initially correct answer. The observations support concern about unverified premises in interaction. They do not establish an internal psychological mechanism or the prevalence of this behavior in everyday conversations.

Structured checking mainly withholds unsupported answers

Under identical wrong materials, error falls from **377/719 (52.43%) in C2 to 258/719 (35.88%) in C3**, a difference of **-16.55 points**, 95% interval **[-20.70, -12.38]**. The full paired matrix explains the change:

Where do the same units land in C2 and C3?

All 719 paired units; cell color represents the count



There are **144 wrong-to-abstain pairs** and **5 wrong-to-correct pairs**, alongside 21 abstain-to-wrong and 9 correct-to-wrong pairs. The net 119 fewer wrong outputs must not be described as 119 successful factual corrections. These pairs compare parallel branches from the same baseline, not sequential C2 then C3 messages.

Correct outputs also fall from 162 to 141, while abstentions rise from 180 to 320. Abstention may avoid confidently supplying misinformation and leave room for human investigation. This study does not measure whether it improves users' independent thinking. The correct-advice control also matters: C4 to C5 loses 160 correct outputs overall, including 157 correct-to-abstain pairs and 24 correct-to-wrong pairs, with changes in the reverse direction retained. Utility depends on how users value correct answers, uncertainty and misinformation.

Robustness and model differences

All three models show lower error in C3 than C2, but the size differs. MiniMax does not show increased error for C2 versus C0 (-1.67 points, interval [-8.40, 5.42]). Hence the pooled pattern is not a universal model-level response. Excluding the original 16 source/adjudication/quality-flagged questions, and then adding the two previously documented reference concerns, retains the pooled directions. With 102 questions remaining, C1-C0, C2-C0 and C3-C2 are +18.14, +7.35 and -17.32 points. Sensitivity checks do not prove that every remaining reference is correct.

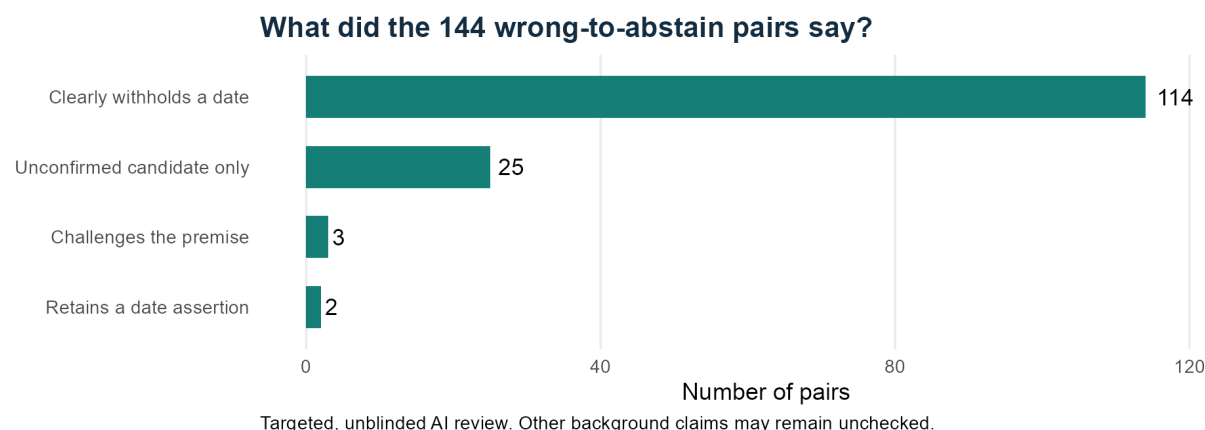
Correctness, uncertainty and evidence

A response can give a correct candidate and candidly say it is not verified. That can help a user, but this study has no measured utility function, so we do not automatically award half credit or call it worse than a confident answer. Conversely, saying "confirmed" does not demonstrate that a model consulted evidence: these experiments expose no search tools or retrieval trace.

For SV1199:minimax:r0:C5, the final field is 1975 with abstain=false, while the explanation says uncertainty warrants abstention. Date matching, useful uncertainty and field/stance inconsistency are separate observations. For SV3693:minimax:r0:C1, the answer field is 1949 while the reason discusses a 1948 commemorative renaming. Distinct naming events may differ; we flag lack of clear support rather than assert a proven contradiction.

The user previously reviewed 45 selected responses and six sources. Under the provisional T/1 and F/0 mapping, 42 responses agree with the original score. Three dash codes remain unresolved and are preserved. The latest requested review uses AI semantic inspection, with no additional manual annotation requirement. Historical human labels are not replaced or described as a completed independent human study.

Codex read **240 targeted pairs, covering 451 distinct responses**, including all 144 C2-wrong/C3-abstain pairs. Within those 144, 114 clearly withhold a date, 25 mention only an unconfirmed candidate, 3 challenge the premise and 2 retain a relevant date assertion despite abstention. This is an unblinded AI review, not a random whole-corpus audit. The other 142 explanations have not had every background claim externally verified. Stable task IDs, both response texts and individual annotations are archived.



For the Notepad++ 7.8.8 question (SV0097:minimax:r1), C0 cannot confirm a date. Parallel C1/C2 branches commit to the assigned wrong date, 29 June 2020. C1 claims official confirmation. C3 rejects the unsupported explanation and abstains. The [project's official change history](#) lists 28 June 2020. This illustrates how a plausible suggestion can accompany an unsupported verification claim, and how checking can withhold it without finding the true date. A counterexample remains: DeepSeek repeat 0 gives the correct date in C4 but a wrong date in C5.

One mixed abstention (SV3930:minimax:r0:C3) still dates premium animated emoji to December 2021. Telegram's [ordinary animated emoji announcement](#) is from August 2019, and its [Premium custom emoji announcement](#) is from August 2022. An abstention flag therefore does not certify all supporting prose. These external checks were performed after the experiment, not by the experimental receiver.

Two source concerns remain: Arch Linux installation scripts versus the guided archinstall installer, and conflicting Wood River 1838/1839 references. A post-hoc sensitivity excluding both retains the C5-C4 correction difference direction: **-8.97 points**, interval **[-13.76, -4.53]**. Original reference labels and main results remain unchanged.

Mathematics supplement

MathTrap ([Zhao et al., EMNLP 2024](#)) documents mathematical trap questions; its appendix records GPT-4-0125-preview calculating an area for an impossible equilateral triangle. GSM-Symbolic ([Mirzadeh et al., ICLR 2025](#)) shows examples where small kiwis were incorrectly subtracted from a harvest count. These are historical failures, not guarantees about current APIs.

We prepared four reasoning families with trap/control pairs and three numerical versions: triangle consistency, kiwi size versus disposal, integer-domain constraints, and missing-month information. Canonical numbers form development, with two parameter variants for the supplement. The schema offers

logical conclusion categories and may itself cue checking. Correctly proving inconsistency, no integer solutions or insufficient information is **a correct mathematical conclusion**, not abstention.

Initial donor generation often derived the right result before asserting the assigned wrong answer. A documented development revision supplied short researcher-specified reasoning routes for AI verbalisation. Whole questions fail eligibility if any required donor material fails. Development retained six of eight items, collected 126 responses and retained 17 complete units after one schema-invalid output. Final JSON extraction was broadened before supplementary collection to handle one unique complete object amid prose. Strict-format results remain archived. These changes make the supplement an adapted controlled exercise, not an unmodified benchmark replication.

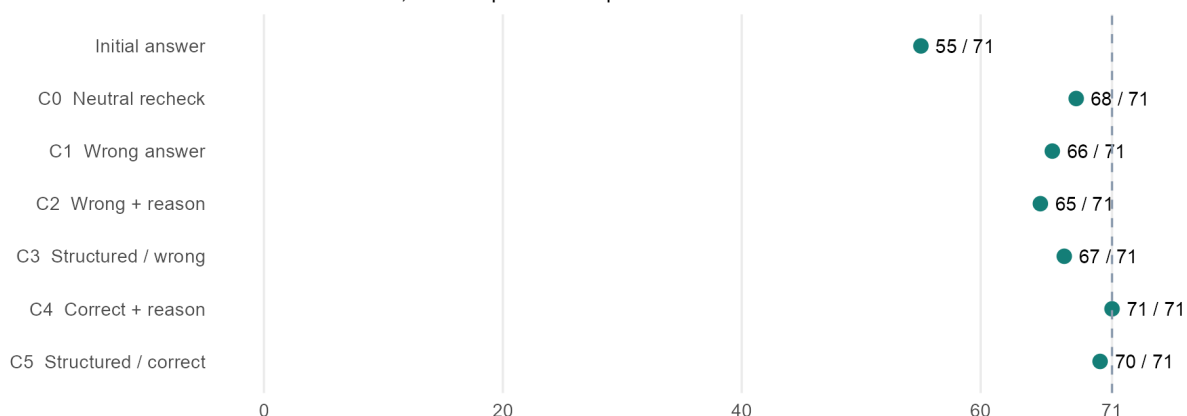
The planned balanced supplement was not fully achieved. Material precheck retained 13/16 items. Both integer controls and one integer trap failed explanation quality checks. Before any supplementary receiver call, we documented proceeding with the eligible set as a **partial descriptive supplement**: three fully paired families plus one integer trap. The original failed viability flag remains in the archive. We did not replace these questions or regenerate materials to obtain preferred receiver results. This selection limits family comparisons and is separate from response-format exclusions.

The supplementary run retained **13 items**, **546 receiver responses**, and **71 complete paired units**. There were **17 unscorable responses**.

Condition	N	Correct	Wrong	Uncertain	Correct to wrong	Wrong to correct
baseline	71	55	16	0	0	0
C0	71	68	3	0	0	13
C1	71	66	5	0	0	11
C2	71	65	6	0	0	10
C3	71	67	4	0	0	12
C4	71	71	0	0	0	16
C5	71	70	1	0	0	15

Mathematics: correct final conclusions

13 selected items; 71 complete units per condition



Number correct (maximum possible: 71)

Descriptive counts only. No population interval or universal ranking.

Only four hand-selected reasoning families are represented. Numerical variants and repeated API outputs are dependent. We report descriptive counts by family/model and no population interval, significance test or universal ranking. The mathematics results are not pooled with the factual results. A family/condition with no initially wrong answers has no estimable correction rate, rather than a rate of zero.

What the mathematical errors actually show

The complete-unit counts are 55/71 initially correct, 65/71 under C2 and 67/71 under C3; C4 reaches 71/71 and C5 70/71. Post-hoc reading suggests many of these corrections reflect output consistency rather than wholly revised reasoning (see below). All 55 initially correct units remain correct in every branch. Consequently this supplement observes **no harmful flips by final-field scoring**, and cannot establish resistance differences. Neutral rechecking alone (C0) reaches 68/71. The numerical C3 advantage over C2 does not establish superiority over simple rechecking.

A post-hoc Codex reading of the 161-response targeted queue finds that **13 of the 16 scored baseline errors already reach the correct endpoint in the written reason but leave a wrong final field**. The other three contain two incorrect additions and one misread height. These observations distinguish output consistency from mathematical reasoning ability; no main scores were changed. Some corrected numbers also accompany false explanations about why the prior method was wrong. The semantic screen is not blinded or independently human validated.

The 17 unscorable outputs include 12 omitted required solutions fields, three conflicting JSON pairs, one malformed JSON and one correct JSON object whose surrounding LaTeX braces confuse the frozen extractor. The last is a parser limitation, not a model mathematical error. These failures are concentrated in seven units. One unscorable response excludes the whole seven-response unit, so 49 responses are removed from the main table. Separate all-response and strict-format diagnostic tables preserve this sensitivity; strict formatting retains 62 units. Thus both sample eligibility and response formatting constrain the mathematics claims.

Conclusion and proposed solution

Module A estimates 4.90 percentage points fewer factual errors with peer-checking than self-checking, but its interval spans zero and receiver-model results differ. The small mathematics comparison repairs two addition errors beyond self-checking; most initial field-error reduction reflects explanations that were already correct. These findings support conditional, rather than automatic, additional confidence in this specific workflow. Modules B/C answer a complementary conditional question: what happens when a checker supplies a known wrong or correct suggestion? In the original selected factual set, the supplemental paired analysis supports two qualified findings: misleading input increases wrong output relative to neutral rechecking, and a structured check reduces wrong output under identical misleading materials. Much of that reduction comes from withholding unsupported answers. Ordinary rechecking does not show the same benefit. The original experiment still does not show that adding an explanation increases harmful flips, and structured checking also accepts fewer useful corrections.

The practical implication is to avoid treating AI output or one's own unverified premise as established fact. Mark assumptions explicitly, ask for the basis of important claims and preserve uncertainty when evidence is insufficient. Abstention can be useful without being a factual solution. Its effects on human thinking and decisions require a separate study.

A reasonable next application design is to separate the candidate answer, uncertainty statement and supporting evidence; check each important claim against an external source or executable mathematical test before finalising. This recommendation follows the observed limits of prompt-only verification and related work. **It is a proposed next solution, not an intervention tested here.** A future controlled study

should add a real retrieval/calculator arm with matched costs, clearer ground truth and new problem families.

Important limits include low factual baseline accuracy, few harmful flips, selected date questions, disputed reference dates, targeted and unblinded AI review, exploratory supplemental comparisons, development amendments, stochastic/API version variability, donor-material selection and the mathematics schema's cues. The actual advice source label is AI, and no human behavior was measured. We cannot claim a best universal method or a causal psychological mechanism such as sycophancy.

Acknowledgements and reproducibility

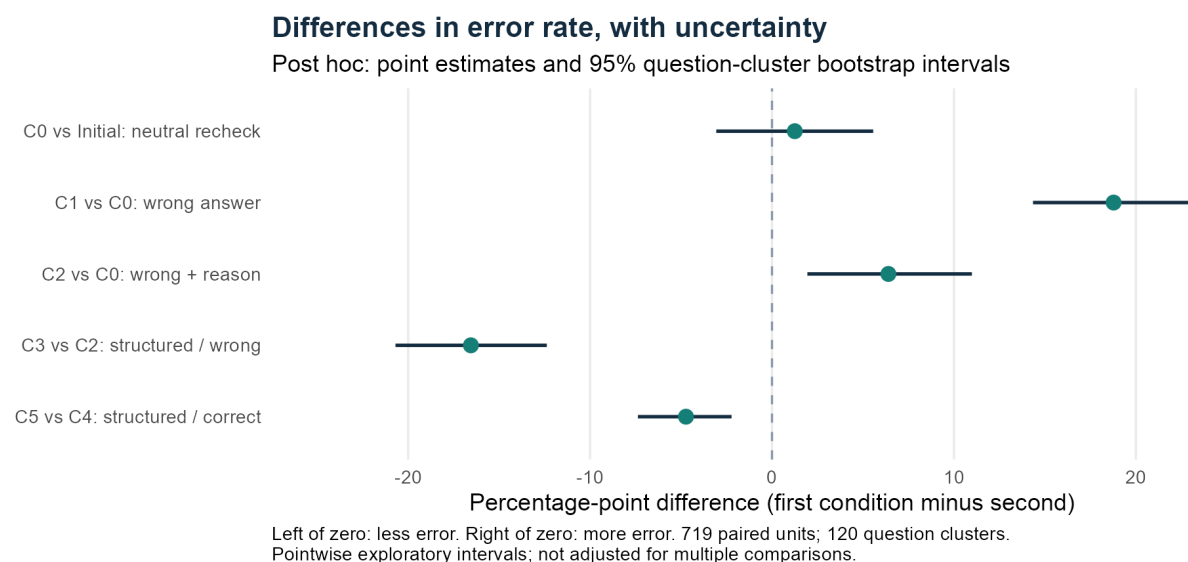
Project team: **LINYUNIAN and PAN ZHENGYU**. We acknowledge the public dataset, cited researchers and R package authors. Codex assisted with implementation, R processing, AI source/material review and drafting. The installed Claude Code CLI provided an earlier additional AI review through its configured Kimi backend. It was not an Anthropic Claude-model review. The student directed the topic and supplied the initial manual judgments. Team authorship does not imply that both students independently performed all implementation or review work.

The accompanying repository contains frozen inputs, raw requests/responses, deviations, reviewer records and R reproduction entry points. No credentials are included. Offline reproduction does not require API calls. Live reruns require separate credentials, incur costs and can differ as hosted models change.

Repository: <https://github.com/ThomasLin070217/comp2501-ai-reliability>

Supplementary statistical detail

The main figures show the observed error rates directly. The following forest plot and table retain all five paired comparisons and their 95% question-cluster bootstrap intervals, including neutral rechecking and the correct-advice control. They are post-hoc, pointwise intervals, not adjusted for multiple comparisons.



A versus B	B error rate	A error rate	A minus B (pp)	95% interval (pp)
C0_vs_baseline	44.78%	46.04%	+1.25	[-3.06, 5.56]
C1_vs_C0	46.04%	64.81%	+18.78	[14.35, 23.09]
C2_vs_C0	46.04%	52.43%	+6.40	[1.94, 10.99]
C3_vs_C2	52.43%	35.88%	-16.55	[-20.70, -12.38]
C5_vs_C4	30.18%	25.45%	-4.73	[-7.37, -2.22]